

ElectroEncephaloGraphy Verification of Speed Adaptation in Time-distortion Ski Training System

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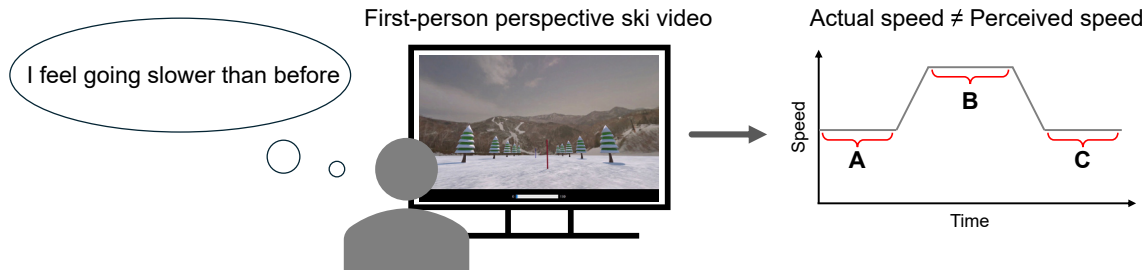


Figure 1: When people watch a first-person perspective ski video that increases speed and then returns to the original speed, people feel going slower after returned speed than before the increased speed, although actually the same speed.

ABSTRACT

Speed adaptation is a perceptual effect in which continuous exposure to fast motion causes subsequently viewed motion to appear slower. Although previous works show that skill training with time distortion in VR is effective, the statistical evidence underlying how they work remains limited. In the gradual speed-increase condition, non-beginner participants reported significantly lower subjective speed. EEG results from consistent responders showed significant changes in parietal Alpha (Pz) and posterior temporal Theta (P8) bands. These findings confirm that the VR time distortion system induces measurable *speed adaptation*, accompanied by corresponding changes in parietal and posterior temporal neural activity.

Index Terms: EEG, Speed Perception, Speed Adaptation.

1 INTRODUCTION

Our subjective perception of time and speed often diverges markedly from physical reality. A classic example is the *speed adaptation* phenomenon: after driving at high speeds on a highway, cars on a normal road appear unusually slow. It is not because they have decelerated, but because one's internal reference for "normal" speed has temporarily shifted [20]. This perceptual recalibration is fundamental to human speed estimation.

Time distortion techniques in Virtual Reality (VR) training deliberately exploit this principle. Prior work has shown that slowing down virtual motion can enhance real game performance in fast motor tasks such as racket sports [26, 7], while accelerating visual motion in VR skiing can make standard speeds feel slower after training, thereby improving downhill control [10]. Although

these studies demonstrate behavioral benefits, the underlying perceptual changes remain unclear. In particular, it is unknown how users' subjective speed perception evolves during exposure to distorted motion and whether such changes manifest in corresponding neural activity. Neuroscience research suggests that motion perception and speed processing are closely tied to neural dynamics in occipital and temporal regions, especially around the area MT/V5. Variations in perceived motion speed have been associated with systematic power changes in Electroencephalography (EEG) signals [9, 8, 19]. However, despite the widespread use of time distortion in VR training, no prior work has examined whether speed adaptation induced by time distortion produces measurable EEG signatures.

Looking forward, a major goal of VR-based motor training is to develop scientific and controllable time distortion strategies that can be adapted to individual users. A principled understanding of how time distortion alters subjective speed and neural processing would allow future VR systems to dynamically adjust visual timing in response to users' perceptual state, thereby enabling personalized and more effective training programs. As a concrete step toward this broader objective, we focus on a VR time distorted skiing scenario, a representative task where visual speed plays a critical role in performance and where time distortion has been shown to yield measurable training benefits.

To address this gap, we investigated how time distorted first-person skiing videos influence both subjective speed perception and neural responses. Participants provided continuous speed ratings while their EEG was recorded. Our analysis shows that non-beginner participants exposed to a gradual speed-increase stimulus exhibited significant downward shifts in perceived speed. Furthermore, EEG results from participants with stable reporting patterns revealed significant modulation in parietal Alpha (Pz) and posterior temporal Theta (P8) activity, suggesting neural correlates of speed adaptation.

Our contributions are summarized as follows:

- **Joint Perceptual&Neural Dataset:** We collected synchronized subjective speed ratings and EEG signals during exposure to time distorted first-person ski videos, enabling quantitative analysis of a perceptual phenomenon.

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- **Behavioral Evidence of Speed Adaptation:** We demonstrate that a gradual speed-increase manipulation induces clear speed adaptation effects in non-beginner participants.
- **Neural Correlates of Adaptation:** We provide preliminary EEG evidence that speed adaptation modulates neural activity in parietal and posterior temporal regions.

2 RELATED WORK

2.1 Time distortion System

To improve the effectiveness of training, various training systems have been developed, such as systems that utilize video analysis [25], systems that utilize VR [6, 23], and systems that utilize electrical stimulation [24, 17]. A time distortion training system has been developed that is effective for beginners who find it difficult to master skills at normal speeds, and for advanced players who want to improve the effectiveness of their training for real-world competitions and other events. Time distortion is a technique that allows a virtual environment to correspond to the speed of real-world objects by making the speed of objects and movements faster or slower than the speed people are used to or the actual speed. For example of sports time distortion system, Virtual Reality (VR) training system for returning high-speed spin serves in table tennis [26], VR training system that utilizes time distortion to improve skiing skills [10], VR tennis training system that synchronize the ball's time with the user's physical movement [7].

This approach is increasingly being applied to sports where manipulating the speed of objects in the real world is difficult or impractical. However, there is no knowledge of how fast or slow the movement and object should be to provide effective training in these research. Therefore, the training effects of varying speed conditions in these time distortion system should be investigated.

2.2 Speed Perception

Smith et al. reported that adaptation to a specific velocity reduces the subsequently perceived speed [20]. Their experiments demonstrated that, depending on the specific adaptation velocity and conditions, this effect is highly variable: perceived speed could be reduced to as low as 40% of the original, or conversely, be elevated to as high as 150%. Other foundational work, such as that by Denton et al., has investigated subjective speed perception using magnitude estimation methods. For example, they instructed participants to make judgments based on halving or doubling a constant reference speed [4]. Further previous works of speed perception include those that examined the speed change required to recognize an increase or decrease in a vehicle's speed [22], those that examined the effect of speed adaptation on speed judgments [18], and those that altered speed perception using external stimuli [21, 1].

These studies have not clarified the conditions necessary to effectively induce speed adaptation, nor have they provided corroborating EEG evidence.

2.3 EEG about Speed Perception

There are many studies in the field of the EEG, such as estimation of emotions from brain waves [12, 11], brain-computer interfaces [16, 13, 5], and changes in brain activity due to external stimuli [15, 3], and EEG related to speed is investigated. In a study by Shen et al., EEG and behavioral data were recorded from participants while the speed was varied for each trial in an experiment to examine the perception of disappearing moving object. [19] The results revealed a detected Event-Related Potential (ERP) that was concentrated at the occipital electrode PO3. In an experiment conducted by Chawla et al., investigated the optimal speed for responses in area V5, it was found that area V5 responds more sensitively to speed than area V1. It has been shown that applying TMS to area V5 in a

speed estimation task after adapting to the object's speed results in a perceived slower speed [2].

These studies have reported that tasks involving speed changes elicit brain activity in the occipital region, where the V5/MT area is located. However, most of these experiments were conducted using visual stimuli such as random dots or grid patterns, which are far removed from real-world situations.

3 METHOD

3.1 Joint Perceptual&Neural Dataset

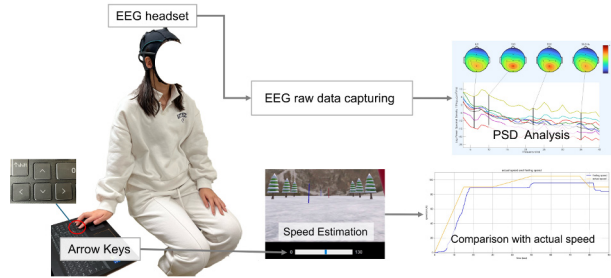


Figure 2: Overview of data recording method. Captured EEG data from EEG headset is used analysis with power spectral density and captured subjective speed from arrow keys on laptop is used comparison with actual speed.

To investigate effect of speed adaptation, we conducted an experiment to record both subjective speed and EEG. EEG raw data which captured by EEG headset participants wearing is analyzed with power spectral density (PSD) after the preprocess. Subjective speed is recorded using arrow keys, afterwards compare with actual speed. For the experimental stimuli, we utilized time distorted first-person skiing videos that can simulate conditions inducing speed adaptation artificially.

We propose the following research questions: **RQ1:** Returning to original speed after adaptation to higher speed will cause a decrease in subjective speed, making it feel slower than its original baseline. **RQ2:** Changes in subjective speed related with increasing or decreasing neural activity in around the posterior temporal cortices corresponding to Area V5.

3.2 Procedure

The overall experimental procedure is illustrated in Figure 3.

First, participants completed a pre-questionnaire via Google Forms, which included questions about their skiing proficiency. After that, EEG headset was fitted.

Next, participants memorized the movement speed according to the interval of the pole flowing by watching three sample videos. After watching the sample videos, a practice session was provided, allowing participants to learn how to adjust the speed bar using the laptop's arrow keys. Once participants understood how to adjust the speed bar, the experiment started.

Each trial began with the start of EEG recording. Ten seconds after the EEG recording began, the test video play and the recording of subjective speed started at the same time. Participants were instructed to continuously predict actual speed during watching video and to adjust the speed bar immediately whenever they perceived a speed change. After watching the test video, participants completed a questionnaire regarding the speed they perceived during that specific trial. Procedure from start EEG recording to complete questionnaire was repeated for all three test videos. After all trials were completed, participants filled out a post-questionnaire about their overall subjective speed throughout all trials, as far as they remember.

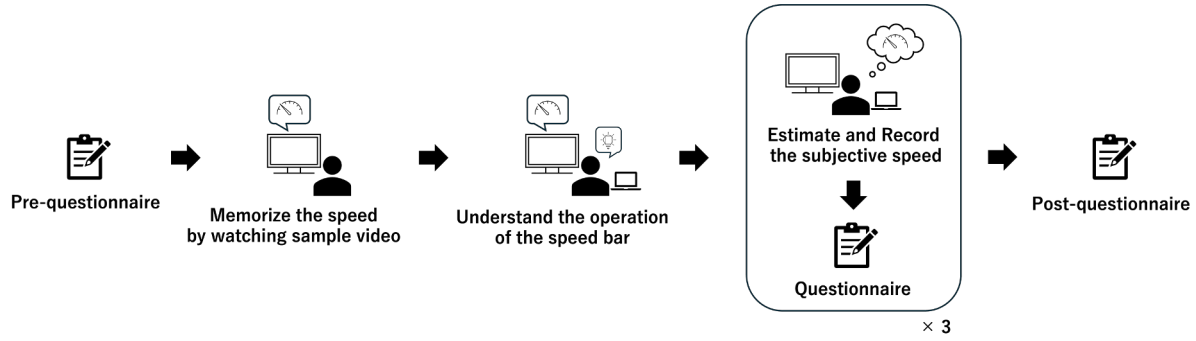


Figure 3: Experimental procedure. Speed estimation task was performed for each of the three test videos.

3.3 Video

To make participants memorize the speed, we prepared three 40-second sample videos that the speed bar located at the bottom of the video moved automatically to indicate the actual speed. Although the maximum speed in the test video was 120 km/h, considering the possibility that the subjective speed may be faster than the actual speed, the maximum speed in the sample video was set to 130 km/h, and the maximum value of the speed bar was also set to 130 in both videos. Regarding the speed changes in the sample videos, we decided to provide both patterns where the speed changes rapidly or slowly within a single video, as well as patterns where the acceleration changes rapidly or slowly, and to ensure that one of the videos would reach maximum speed of 130 km/h.

The main experiment consisted of three 90-second test video conditions: **S1**: A constant speed with no speed change. **S2**: Increase speed from 80 km/h to 120 km/h in 20 seconds, then return to 80 km/h in 5 seconds. **S3**: Increase speed from 90 km/h to 105 km/h in 20 seconds, then return to 90 km/h in 5 seconds.

4 EXPERIMENT

4.1 Participants

A total of 12 participants (8 male, 4 female) took part in the experiment. The pre-questionnaire collected gender, age, skiing proficiency and skiing experience in the last 5 years. The mean age of the participants was 24.58 years ($SD = 4.07$). Based on a pre-questionnaire about their skiing proficiency, the group was categorized as follows: 5 beginners (no experience or unable to ski), 4 intermediates (able to ski in a V-shape, or "snowplow"), and 3 advanced skiers (able to ski parallel).

4.2 Set up

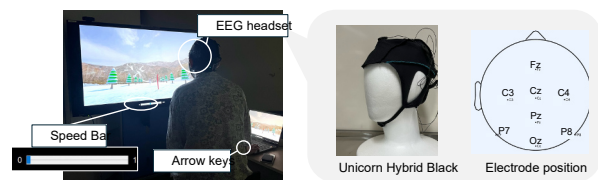


Figure 4: Scene of the experiment. The experiment was conducted in a dark room, with the participant sitting in a chair one meter from the screen.

The experimental setup is shown in Figure 4. The distance between the participant and the screen was set to 1 meter, and the experiment was conducted in a dark room to make the screen easier

to see. As cautions during the experiment, participants were instructed to keep their heads as steady as possible their brain waves were being recorded, and to pay attention to the pole while watching the video in order to ensure they remained concentrate on the task.

For EEG acquisition, we used the Unicorn Hybrid Black¹, a dry-electrode recording system. EEG data were recorded from 8 electrodes placed at Fz, C3, Cz, C4, Pz, P7, Oz, and P8, according to the international 10/20 system. Reference and ground electrodes were attached to the mastoids (under each ear). The EEG signals were recorded at a sampling frequency of 250 Hz. This dry-electrode method was selected as Matthew et al. have previously demonstrated that reliable measurements can be obtained without conductive gel [14].

The recording of subjective speed were using the arrow keys on a laptop. All video stimuli were displayed on a 60-inch monitor.

4.3 EEG Data Analysis

EEG data analysis was performed using MATLAB R2025a. The preprocessing pipeline consisted of: 1) applying a band-pass filter (0.5–40 Hz) to the raw data; 2) performing Independent Component Analysis (ICA) using reference; 3) identifying and rejecting components labeled other than brain labels based on IC Label. and 4) re-referencing the data. Following preprocessing, Power spectral density for each frequency at each electrode were Z-score normalized, and the band power for each frequency bands was calculated. Power spectral density for each frequency at each electrode were extracted using the EEGLAB.

To facilitate the analysis, we segmented the test video into three key periods based on the actual speed: Part A (15–30 s), Part B (50–75 s), and Part C (80–90 s).

The analysis windows for both subjective speed and EEG were determined as follows. For the subjective speed, we primarily defined the evaluation window as the period from the point at which the participant's subjective speed stabilized until the end of that part. If a participant's subjective speed failed to stabilize, the entire duration of the part was used for the evaluation.

For the EEG, we defined default 5 seconds analysis windows for each part: Part A (25–30 s), Part B (70–75 s), and Part C (84–89 s). The window for Part C was set to end 1 second before the video's conclusion to prevent contamination from artifacts related to the video's offset. However, two exception were applied. First, if the participant's stable subjective speed window differed from the default EEG window, the EEG analysis was performed using the time range derived from the stable subjective speed. Second, if a participant's subjective speed failed to stabilize during Part A or Part C

¹<https://www.gtec.at/product/unicorn-hybrid-black/>

that specific trial was excluded from the EEG analysis entirely, as a reliable EEG window could not be determined.

To test our hypotheses, we conducted t-tests on the differences in band power between the each parts. Furthermore, we performed a correlation analysis to investigate the relationship between the change in subjective speed and change in band power.

5 RESULT

5.1 Subjective Speed

A paired-samples t-test was conducted to compare the subjective speed ratings between Part A and Part C for all 12 participants. This initial analysis revealed no significant difference for either the S2 condition ($t = -0.3787$, $p = 0.7121$) or the S3 condition ($t = 1.7410$, $p = 0.1095$).

Because the responses of beginner participants were likely to introduce a strong bias and lacked reliability for assessing speed perception, we excluded 5 beginner data from subsequent analyses and conducted a paired-samples t-test for 7 intermediate and advanced participants. This analysis result is that the S2 condition remained non-significant, whereas the S3 condition showed a significant effect ($t = 2.6881$, $p = 0.0361$).

Examples of subjective speed ratings for individual participants are shown in Figure 5. As illustrated by P3 (Figure 5, right), repeated exposure to accelerated motion successfully induced speed adaptation, causing the actual speed to feel slower over time. P2's result (Figure 5, left) shows that the subjective speed remained nearly constant despite the gradual increase in actual speed. A noticeable drop in subjective speed occurred only when the actual speed decreased, leading to a clear reduction in Part C compared to the baseline. Conversely, two participants in the S2 condition reported higher subjective speed in Part C than in Part A. This pattern suggests that strong acceleration in the time distortion condition may fail to induce speed adaptation and may instead lead some participants to perceive the motion as faster than the original speed.

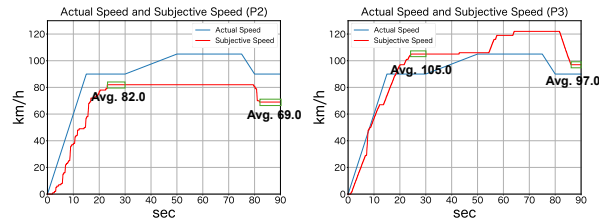


Figure 5: The example of graph about comparison between subjective speed and actual speed in S3 condition.

5.2 EEG

Based on the behavioral results, our EEG analysis focused solely on the S3 condition, as this was the only condition showing a significant change in subjective speed.

The EEG data was selected as follows. Among the 12 participants, 5 were excluded because they were classified as beginners. An additional 2 participants (P4 and P9) were removed because their subjective speed did not stabilize during either Part A or Part C. Thus, the EEG analysis was conducted on the remaining five participants. A paired-samples t-test on 5 participants' subjective speed ratings between Part A and Part C confirmed a significant behavioral effect of speed adaptation ($t = 6.4781$, $p = 0.0029$), indicating this subset reliably exhibited the perceptual shift required for neural comparison.

We then examined whether this behavioral change was accompanied by corresponding neural modulation. For each electrode, paired-samples t-tests were performed on band-power differences

(Part C minus Part A), and the False Discovery Rate (FDR) procedure was applied to correct for multiple comparisons. Significant effects emerged in two regions: Pz-Alpha ($t = 10.0010$, $p = 0.0384$) and P8-Theta ($t = 9.2523$, $p = 0.0384$). These two sites lie in parietal and posterior temporal areas often associated with motion processing and visuospatial integration. This result suggests that speed adaptation may recruit neural mechanisms related to attention reallocation and motion-speed estimation.

Overall, these findings indicate that EEG is sensitive to perceptual changes induced by time distorted stimuli, supporting the potential for future systems to monitor or adapt time distortion based on neural feedback.

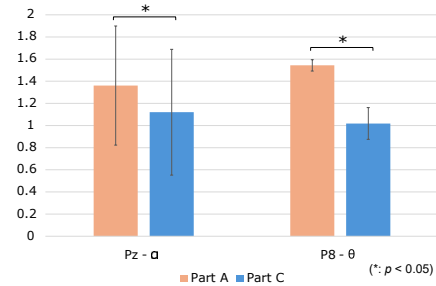


Figure 6: The band power range difference between Part A and Part C for Pz-Alpha and P8-Theta.

6 DISCUSSION

We further examined whether trial-by-trial fluctuations in subjective speed were reflected in corresponding changes in neural activity. For each electrode, we computed the correlation between the difference in subjective speed (Part C minus Part A) and the difference in band power over the same interval. Part C was segmented into successive 1-second windows with a 0.25-second step (e.g., 85–86 s, 85.25–86.25 s, ...). For each window, we calculated the deviation from the participant's mean subjective speed and mean band power in Part A. These paired differences were then used to calculate correlation coefficients, with subjective speed and band power.

Analysis results were concentrated higher frequency bands around the posterior temporal cortices and the vertex. Cz-Beta ($r = 0.7143$), Cz-Gamma ($r = 0.7241$), P7-Beta ($r = 0.7952$) are electrode-frequency pairs with correlation coefficients exceeding $r > 0.6$: Pz-Theta ($r = 0.6774$), P7-Gamma ($r = 0.6894$), P8-Beta ($r = 0.6135$), and P8-Gamma ($r = 0.6119$).

Overall, these results suggest a consistent relationship between subjective speed differences and oscillatory activity in the beta and gamma bands, particularly in regions involved in human's visual integration and motion processing.

7 CONCLUSION

This study investigated speed adaptation induced by a time distorted VR skiing system using simultaneously recorded subjective speed ratings and EEG signals. Our results showed that time distortion led clear speed adaptation effects in non-beginner participants. Moreover, changes in subjective speed were associated with modulation in parietal Alpha (Pz) and posterior temporal Theta (P8) activity, suggesting that EEG can capture neural correlates of perceived speed.

These findings highlight the feasibility of using EEG to track differences in subjective speed, and indicate potential applications in real-time BCI systems that investigate suitable speed for training with time distortion system.

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